

# Nguyen Thanh Long

## Full-Stack Engineer

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### SUMMARY

Full-stack engineer, 2.5 years, working in TypeScript on both ends of a production IoT platform: the Next.js dashboard partners use and the NestJS APIs behind it. I own that platform's cloud side end to end, so the same work also covers the database, the message pipeline and the deployments.

### EXPERIENCE

#### Full-Stack Engineer at Rogo Solutions JSC Jan 2024 to present · Hanoi

Own the cloud side of a multi-tenant IoT platform end to end: the partner dashboard, the API, the message pipeline, and the deployments. It runs at a few million API requests a day, with around 15,000 concurrent MQTT connections and roughly 12 million device messages a day.

- Build the partner dashboard in **Next.js** and TypeScript, with **React Query** for server state, Ant Design and Tailwind for the interface, and internationalisation. A single deployment serves multiple partner projects, configurable per project.
- Attribute-based authorisation** in which the permission set is encoded into a compact **custom claim on the JWT**, so a request is authorised from the token itself instead of a database lookup on every call. The React app derives the same rules through **CASL**, so the interface and the API agree on what is allowed.
- Device state streaming on **Kafka** with **MQTT** ingestion: a consumer that coalesces repeated updates into batched writes and commits offsets after the write, autoscaled by **KEDA** on consumer group lag.
- Run the platform on **Kubernetes** and **AWS ECS**, with **Prometheus**, **Loki** and **Grafana** for metrics and logs, and the staging to production path on GitHub Actions, GitLab CI and **ArgoCD**.

### SELECTED PROJECTS

#### Educystal (Next.js, Supabase, Tailwind, PWA) · educystal.vercel.app

A guided crystal-growing app for children: step-by-step programs, a private photo diary, and a gallery a moderator approves. Access rules live in Postgres row level security rather than in the application, so a route cannot leak another child's diary. Installable as a PWA. Designed and built solo.

#### IoT Cloud MCP (TypeScript, NestJS, MCP SDK, Redis)

An MCP server for IoT cloud platforms, so an assistant can query and control devices through one interface rather than a separate integration per platform.

#### Novel Studio (NestJS, PostgreSQL with pgvector)

A long-form writing tool. The awkward part is that prompt cost grows with the story, so it keeps a canon store, a rolling summary and vector retrieval instead of resending history.

### TECHNICAL SKILLS

|                  |                                                                                        |
|------------------|----------------------------------------------------------------------------------------|
| Frontend         | TypeScript, React, Next.js, Tailwind CSS, Ant Design, React Query, React Native (Expo) |
| Backend          | Node.js, NestJS, Express, REST APIs, WebSockets                                        |
| Data             | PostgreSQL, MongoDB, Redis, MySQL                                                      |
| Infrastructure   | Docker, Kubernetes, AWS (ECS, S3), GitHub Actions                                      |
| Messaging & auth | Kafka, MQTT, Firebase Auth, Keycloak, JWT                                              |

### EDUCATION & ADDITIONAL

#### B.Sc. in Software Engineering, FPT University Graduated 2026

Final-year project: a real-time multiplayer game backend (ASP.NET Core, WebSockets, MySQL) using lag compensation to keep play fair across uneven network conditions.

**Certifications:** Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).

Coach competitive-programming students for national Informatics examinations, weekly since 2024. Freelance backend work alongside full-time employment. English: professional working proficiency.